

Project 4

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```
library(geosphere)
```

```
## Warning: package 'geosphere' was built under R version 4.4.2
```

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
crime_data <- read.csv("C:/Users/phani/Desktop/AIT 580/crime_data.csv")
```

```
print("Structure of the dataset:")
```

```
## [1] "Structure of the dataset:"
```

```
str(crime_data)
```

```
## 'data.frame': 20860 obs. of 25 variables:
```

```
## $ X : num 396920 397792 396519 397824 398477 ...
```

```
## $ Y : num 138793 134654 142279 141377 137709 ...
```

```
## $ CCN : int 24091506 24077884 24062904 24106105 24112374 24051500 24017682 240190...
```

```
## $ REPORT_DAT : chr "2024-06-16 09:01:01+00:00" "2024-05-24 06:16:30+00:00" "2024-04-27 1...
```

```
## $ SHIFT : chr "MIDNIGHT" "MIDNIGHT" "DAY" "DAY" ...
```

```
## $ METHOD : chr "OTHERS" "OTHERS" "OTHERS" "OTHERS" ...
```

```
## $ OFFENSE : chr "MOTOR VEHICLE THEFT" "MOTOR VEHICLE THEFT" "MOTOR VEHICLE THEFT" "TH...
```

```
## $ BLOCK : chr "1500 - 1599 BLOCK OF U STREET NW" "50 - 99 BLOCK OF DISTRICT SQUARE S...
```

```
## $ XBLOCK : num 396920 397792 396519 397824 398477 ...
```

```
## $ YBLOCK : num 138793 134654 142279 141377 137709 ...
```

```
## $ WARD : num 1 6 4 4 2 5 6 5 5 1 ...
```

```
## $ ANC : chr "1B" "6D" "4E" "4C" ...
## $ DISTRICT : num 3 1 4 4 3 5 1 5 5 3 ...
## $ PSA : num 301 103 404 404 308 503 104 504 504 304 ...
## $ NEIGHBORHOOD_CLUSTER : chr "Cluster 6" "Cluster 9" "Cluster 18" "Cluster 18" ...
## $ BLOCK_GROUP : chr "004300 3" "010202 5" "002600 2" "002503 1" ...
## $ CENSUS_TRACT : num 4300 10202 2600 2503 4802 ...
## $ VOTING_PRECINCT : chr "Precinct 22" "Precinct 142" "Precinct 48" "Precinct 47" ...
## $ LATITUDE : num 38.9 38.9 38.9 38.9 38.9 ...
## $ LONGITUDE : num -77 -77 -77 -77 -77 ...
## $ BID : chr "" "SOUTHWEST" "" "" ...
## $ START_DATE : chr "2024-06-16 07:00:00+00:00" "2024-05-24 02:19:00+00:00" "2024-04-27 1
## $ END_DATE : chr "2024-06-16 08:48:00+00:00" "2024-05-24 03:42:00+00:00" "2024-04-27 1
## $ OBJECTID : int 606076223 606076245 606076267 606076268 606076269 606076272 606076280
## $ RESPONSE_TIME_MINUTES : num 108 83 4 0 3 5 58 645 5 300 ...
```

```
print("\nSummary of the dataset:")
```

```
## [1] "\nSummary of the dataset:"
```

```
summary(crime_data)
```

```
##           X           Y           CCN           REPORT_DAT
## Min.      :390362   Min.    :127300   Min.      : 5094764   Length:20860
## 1st Qu.:397229   1st Qu.:136234   1st Qu.:24044938   Class :character
## Median :399027   Median :137850   Median :24090977   Mode  :character
## Mean     :399281   Mean     :137795   Mean      :24104671
## 3rd Qu.:401212   3rd Qu.:139739   3rd Qu.:24131153
## Max.     :407807   Max.      :147365   Max.      :25030974
##
##      SHIFT           METHOD           OFFENSE           BLOCK
## Length:20860   Length:20860   Length:20860   Length:20860
## Class :character   Class :character   Class :character   Class :character
## Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##      XBLOCK           YBLOCK           WARD           ANC
## Min.      :390362   Min.    :127300   Min.      :1.000   Length:20860
## 1st Qu.:397229   1st Qu.:136234   1st Qu.:2.000   Class :character
## Median :399027   Median :137850   Median :5.000   Mode  :character
## Mean     :399281   Mean     :137795   Mean      :4.394
## 3rd Qu.:401212   3rd Qu.:139739   3rd Qu.:6.000
## Max.     :407807   Max.      :147365   Max.      :8.000
##
##      DISTRICT           PSA           NEIGHBORHOOD_CLUSTER   BLOCK_GROUP
## Min.      :1.000   Min.    :101   Length:20860   Length:20860
## 1st Qu.:2.000   1st Qu.:207   Class :character   Class :character
## Median :4.000   Median :308   Mode  :character   Mode  :character
## Mean     :3.689   Mean     :373
## 3rd Qu.:5.000   3rd Qu.:506
## Max.     :7.000   Max.      :708
##
##      NA's      :358
```

```
## CENSUS_TRACT VOTING_PRECINCT LATITUDE LONGITUDE
## Min. : 101 Length:20860 Min. :38.81 Min. : -77.11
## 1st Qu.: 3500 Class :character 1st Qu.:38.89 1st Qu.: -77.03
## Median : 7202 Mode :character Median :38.91 Median : -77.01
## Mean : 7792 Mean :38.91 Mean : -77.01
## 3rd Qu.: 9102 3rd Qu.:38.93 3rd Qu.: -76.99
## Max. :980000 Max. :38.99 Max. : -76.91
## NA's :10
## BID START_DATE END_DATE OBJECTID
## Length:20860 Length:20860 Length:20860 Min. :606076223
## Class :character Class :character Class :character 1st Qu.:606403137
## Mode :character Mode :character Mode :character Median :606430046
## Mean :606463053
## 3rd Qu.:606592839
## Max. :606631015
## RESPONSE_TIME_MINUTES
## Min. : 0
## 1st Qu.: 20
## Median : 60
## Mean : 2314
## 3rd Qu.: 210
## Max. :10100450
##
```

```
print("\nMissing Values in Each Column:")
```

```
## [1] "\nMissing Values in Each Column:"
```

```
print(colSums(is.na(crime_data)))
```

```
## X Y CCN
## 0 0 0
## REPORT_DAT SHIFT METHOD
## 0 0 0
## OFFENSE BLOCK XBLOCK
## 0 0 0
## YBLOCK WARD ANC
## 0 0 0
## DISTRICT PSA NEIGHBORHOOD_CLUSTER
## 0 358 0
## BLOCK_GROUP CENSUS_TRACT VOTING_PRECINCT
## 0 10 0
## LATITUDE LONGITUDE BID
## 0 0 0
## START_DATE END_DATE OBJECTID
## 0 0 0
## RESPONSE_TIME_MINUTES
## 0
```

```
print("\nFirst 5 rows of the dataset:")
```

```
## [1] "\nFirst 5 rows of the dataset:"
```

```
print(head(crime_data))
```

```
##           X           Y           CCN           REPORT_DAT  SHIFT METHOD
## 1 396920.3 138793.2 24091506 2024-06-16 09:01:01+00:00 MIDNIGHT OTHERS
## 2 397791.8 134653.6 24077884 2024-05-24 06:16:30+00:00 MIDNIGHT OTHERS
## 3 396518.9 142278.7 24062904 2024-04-27 16:19:40+00:00      DAY OTHERS
## 4 397823.8 141376.7 24106105 2024-07-11 14:17:39+00:00      DAY OTHERS
## 5 398477.3 137708.6 24112374 2024-07-22 23:39:54+00:00 EVENING OTHERS
## 6 403775.6 139530.1 24051500 2024-04-06 23:33:03+00:00 EVENING OTHERS
##           OFFENSE                               BLOCK  XBLOCK
## 1 MOTOR VEHICLE THEFT          1500 - 1599 BLOCK OF U STREET NW 396920.3
## 2 MOTOR VEHICLE THEFT           50 - 99 BLOCK OF DISTRICT SQUARE SW 397791.8
## 3 MOTOR VEHICLE THEFT    4700 - 4799 BLOCK OF BLAGDEN TERRACE NW 396518.9
## 4           THEFT/OTHER    4000 - 4099 BLOCK OF GEORGIA AVENUE NW 397823.8
## 5           ROBBERY              400 - 499 BLOCK OF N STREET NW 398477.3
## 6           THEFT/OTHER 2470 - 2599 BLOCK OF BALDWIN CRESCENT NE 403775.6
##           YBLOCK WARD ANC DISTRICT PSA NEIGHBORHOOD_CLUSTER BLOCK_GROUP CENSUS_TRACT
## 1 138793.2      1 1B           3 301           Cluster 6      004300 3          4300
## 2 134653.5      6 6D           1 103           Cluster 9      010202 5          10202
## 3 142278.7      4 4E           4 404           Cluster 18     002600 2          2600
## 4 141376.7      4 4C           4 404           Cluster 18     002503 1          2503
## 5 137708.5      2 2G           3 308           Cluster 7      004802 1          4802
## 6 139530.1      5 5C           5 503           Cluster 24     009000 2          9000
##           VOTING_PRECINCT LATITUDE LONGITUDE           BID           START_DATE
## 1      Precinct 22 38.91700 -77.03551           2024-06-16 07:00:00+00:00
## 2      Precinct 142 38.87971 -77.02545 SOUTHWEST 2024-05-24 02:19:00+00:00
## 3      Precinct 48 38.94840 -77.04016           2024-04-27 15:26:00+00:00
## 4      Precinct 47 38.94028 -77.02510           2024-07-11 13:27:00+00:00
## 5      Precinct 18 38.90723 -77.01756           2024-07-22 17:57:00+00:00
## 6      Precinct 139 38.92364 -76.95646           2024-04-06 20:30:00+00:00
##           END_DATE  OBJECTID RESPONSE_TIME_MINUTES
## 1 2024-06-16 08:48:00+00:00 606076223           108
## 2 2024-05-24 03:42:00+00:00 606076245           83
## 3 2024-04-27 15:30:00+00:00 606076267           4
## 4 2024-07-11 13:27:00+00:00 606076268           0
## 5 2024-07-22 18:00:00+00:00 606076269           3
## 6 2024-04-06 20:35:00+00:00 606076272           5
```

```
numeric_columns <- sapply(crime_data, is.numeric)
print("\nSummary Statistics for Numeric Columns:")
```

```
## [1] "\nSummary Statistics for Numeric Columns:"
```

```
print(summary(crime_data[, numeric_columns]))
```

```
##           X           Y           CCN           XBLOCK
## Min. :390362 Min. :127300 Min. : 5094764 Min. :390362
## 1st Qu.:397229 1st Qu.:136234 1st Qu.:24044938 1st Qu.:397229
## Median :399027 Median :137850 Median :24090977 Median :399027
## Mean :399281 Mean :137795 Mean :24104671 Mean :399281
## 3rd Qu.:401212 3rd Qu.:139739 3rd Qu.:24131153 3rd Qu.:401212
## Max. :407807 Max. :147365 Max. :25030974 Max. :407807
```

```
##
##      YBLOCK      WARD      DISTRICT      PSA
##  Min.   :127300   Min.    :1.000   Min.    :1.000   Min.    :101
##  1st Qu.:136234   1st Qu.:2.000   1st Qu.:2.000   1st Qu.:207
##  Median :137850   Median :5.000   Median :4.000   Median :308
##  Mean   :137795   Mean    :4.394   Mean    :3.689   Mean    :373
##  3rd Qu.:139739   3rd Qu.:6.000   3rd Qu.:5.000   3rd Qu.:506
##  Max.    :147365   Max.    :8.000   Max.    :7.000   Max.    :708
##                                     NA's    :358
##  CENSUS_TRACT    LATITUDE    LONGITUDE    OBJECTID
##  Min.    :   101   Min.    :38.81   Min.    :-77.11   Min.    :606076223
##  1st Qu.:  3500   1st Qu.:38.89   1st Qu.: -77.03   1st Qu.:606403137
##  Median :  7202   Median :38.91   Median : -77.01   Median :606430046
##  Mean    :  7792   Mean    :38.91   Mean    : -77.01   Mean    :606463053
##  3rd Qu.:  9102   3rd Qu.:38.93   3rd Qu.: -76.99   3rd Qu.:606592839
##  Max.    :980000   Max.    :38.99   Max.    : -76.91   Max.    :606631015
##  NA's     :10
##  RESPONSE_TIME_MINUTES
##  Min.    :      0
##  1st Qu.:     20
##  Median :     60
##  Mean    :    2314
##  3rd Qu.:     210
##  Max.    :10100450
##
```

```
categorical_columns <- sapply(crime_data, is.factor)
print("\nFrequency Counts for Categorical Columns:")
```

```
## [1] "\nFrequency Counts for Categorical Columns:"
```

```
for (col in names(crime_data)[categorical_columns]) {
  print(paste("\nValue counts for", col, ":"))
  print(table(crime_data[[col]]))
}
```

```
landmarks <- data.frame(
  Landmark = c("White House", "Lincoln Memorial"),
  Latitude = c(38.8977, 38.8893),
  Longitude = c(-77.0365, -77.0502)
)
```

```
crime_data <- crime_data %>%
  mutate(
    WhiteHouse_Distance = distHaversine(
      cbind(LONGITUDE, LATITUDE),
      c(-77.0365, 38.8977)
    ) / 1000,
    LincolnMemorial_Distance = distHaversine(
      cbind(LONGITUDE, LATITUDE),
      c(-77.0502, 38.8893)
    ) / 1000
  )
```

```

)

crime_data <- crime_data %>%
  mutate(
    Near_WhiteHouse = ifelse(WhiteHouse_Distance <= 1, "Within 1km", "Beyond 1km"),
    Near_LincolnMemorial = ifelse(LincolnMemorial_Distance <= 1, "Within 1km", "Beyond 1km")
  )

whitehouse_analysis <- crime_data %>%
  group_by(Near_WhiteHouse) %>%
  summarise(Crime_Count = n())

lincoln_analysis <- crime_data %>%
  group_by(Near_LincolnMemorial) %>%
  summarise(Crime_Count = n())

print("Crimes Near White House:")

## [1] "Crimes Near White House:"

print(whitehouse_analysis)

## # A tibble: 2 x 2
##   Near_WhiteHouse Crime_Count
##   <chr>           <int>
## 1 Beyond 1km      19867
## 2 Within 1km       993

print("Crimes Near Lincoln Memorial:")

## [1] "Crimes Near Lincoln Memorial:"

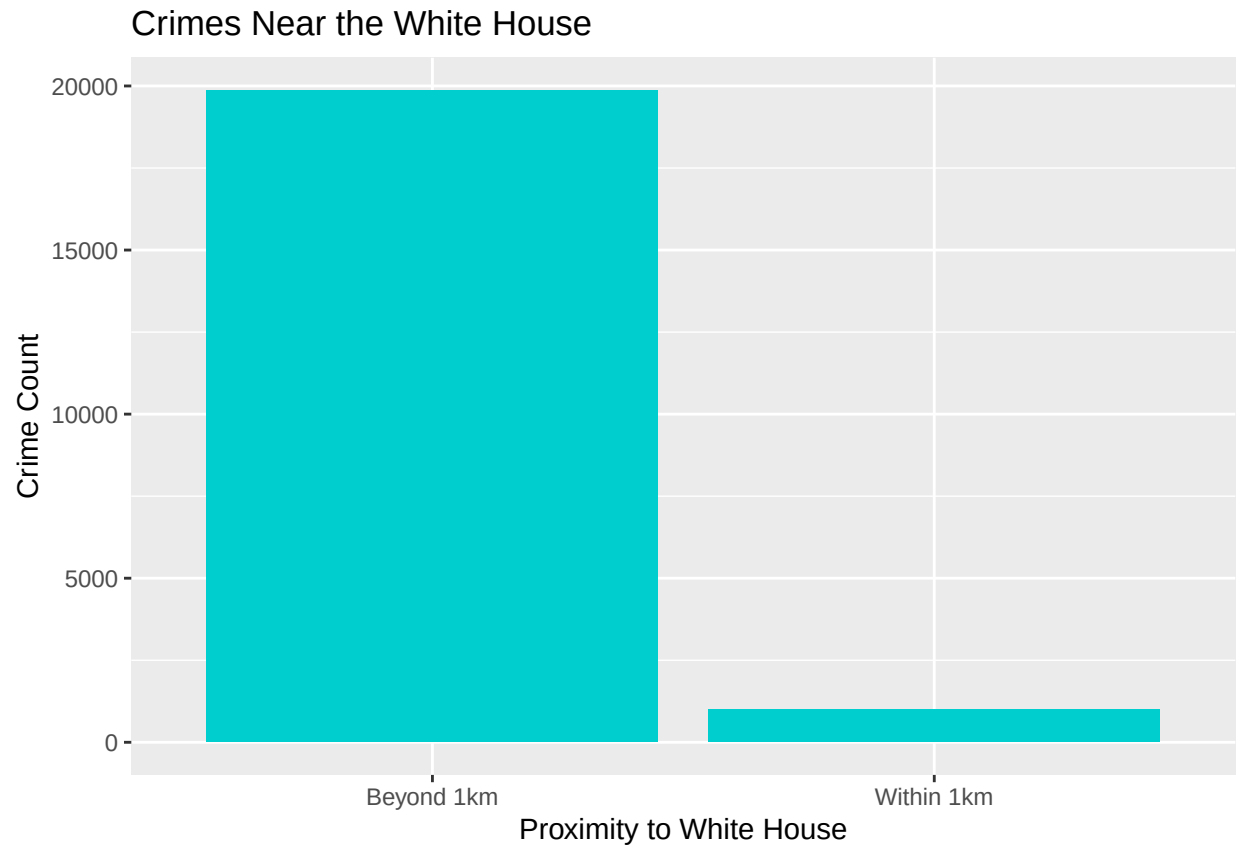
print(lincoln_analysis)

## # A tibble: 2 x 2
##   Near_LincolnMemorial Crime_Count
##   <chr>                 <int>
## 1 Beyond 1km           20801
## 2 Within 1km            59

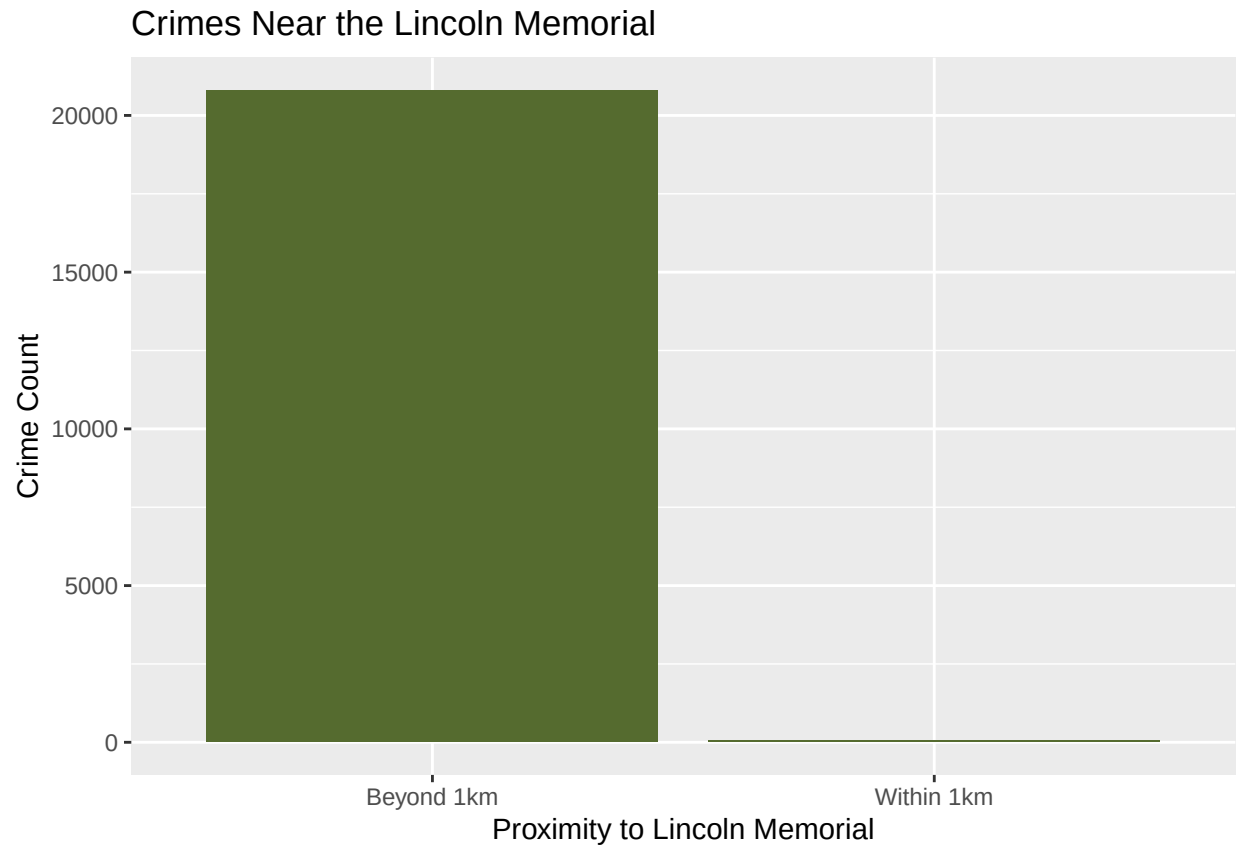
library(ggplot2)

ggplot(crime_data, aes(x = Near_WhiteHouse)) +
  geom_bar(fill = "cyan3") +
  ggtitle("Crimes Near the White House") +
  xlab("Proximity to White House") +
  ylab("Crime Count")

```



```
ggplot(crime_data, aes(x = Near_LincolnMemorial)) +  
  geom_bar(fill = "darkolivegreen") +  
  ggtitle("Crimes Near the Lincoln Memorial") +  
  xlab("Proximity to Lincoln Memorial") +  
  ylab("Crime Count")
```



```
colnames(crime_data)
```

```
## [1] "X"                "Y"
## [3] "CCN"              "REPORT_DAT"
## [5] "SHIFT"            "METHOD"
## [7] "OFFENSE"          "BLOCK"
## [9] "XBLOCK"           "YBLOCK"
## [11] "WARD"             "ANC"
## [13] "DISTRICT"         "PSA"
## [15] "NEIGHBORHOOD_CLUSTER" "BLOCK_GROUP"
## [17] "CENSUS_TRACT"     "VOTING_PRECINCT"
## [19] "LATITUDE"         "LONGITUDE"
## [21] "BID"              "START_DATE"
## [23] "END_DATE"         "OBJECTID"
## [25] "RESPONSE_TIME_MINUTES" "WhiteHouse_Distance"
## [27] "LincolnMemorial_Distance" "Near_WhiteHouse"
## [29] "Near_LincolnMemorial"
```